Ice-driven microplastic oxidation

Editorial assessment, key message, logic arc and a two-figure plan toward *Nature Water*
Source: 5-page figure deck dated 23 October 2026 · prepared 28 October 2026

Bottom line

- One idea in the deck is genuinely new and Nature-family material: **the plastic reshapes the ice that oxidizes it**. Oxidized microplastics (MPs) refine the ice grain structure, and the finer structure is proposed to speed up further oxidation.
- As it stands the deck would not clear a picky *Nature Water* editor. The oxidant source (freeze-made nanobubbles → $^{1}O_2$) is the weakest link. The "positive feedback" is shown only by correlation, and the deck's own decelerating kinetics argue against it. Relevance to natural ice and water is absent.
- What changes that: (i) a discrimination package that rules out four rival ROS sources (Fig. 2a–c); (ii) a causal, bounded test of the feedback (Fig. 2d–f); (iii) a ^{13}C carbon budget (Fig. 2g); (iv) a controlled field deployment with meltwater sampling (Fig. 2h–j).
- Realistic targets today: *Angew. Chem.* or *Nat. Commun.* after (i)–(ii). *Nature Water* only with (iii)–(iv). A presubmission enquiry built around the feedback concept costs little and tests editorial interest.

How to read this report. **Fig. 1** replots the deck's own data, digitized from the PDF (reading error ±2–5%). **Fig. 2** shows the **expected** outcomes of experiments not yet done; every curve comes from a stated model (Section 5.3). Those panels are predictions to design against, not results, and must never be presented as data.

1 Key message and logic arc

1.1 Key message (as the paper should read)

Freezing turns ice grain boundaries into self-reinforcing oxidation reactors for microplastics. Gas expelled by growing ice nucleates nanobubbles on hydrophobic plastic surfaces, where they deliver reactive oxygen. The oxidized, now hydrophilic plastics inhibit ice recrystallization, so each refreeze gives finer ice with more boundary and more bubbles. Freeze–thaw cycling therefore ratchets microplastics toward chain scission, and spring melt flushes the products into surface waters.

冰晶界是微塑料的“自增强氧化反应器”：结冰排出的气体在疏水塑料表面形成纳米气泡并输送活性氧；氧化后变亲水的微塑料抑制冰重结晶，使再结冰的晶粒更细、晶界与气泡更多；冻融循环由此以棘轮方式推动微塑料断链降解，融雪再把产物带入地表水。

Framing by journal: for *Nature Water* lead with the water-systems consequence (what seasonal ice does to plastics, and what meltwater delivers). For *Nat. Commun.*, *JACS* or *Angew.* lead with the mechanism (ice microstructure ↔ interfacial chemistry coupling).

1.2 Logic spine, with evidence status today

Step	Claim	Deck evidence	Status	Closes the gap
S1	Ice, not water, oxidizes C–H polymers; PTFE inert	p1B: XPS O/C, 9 polymers	Supported for PE, PP, PS, PVC; O-containing polymers ambiguous	ΔO/C vs pristine, C1s fitting, FTIR carbonyl index; drop PVA
S2	Freezing exsolves gas into nanobubbles that generate ROS	p2: NTA, DCFH, TEMP–EPR, imaging	**Unproven**: bubble identity and ROS species	Fig. 2a, 2b, 2c
S3	Bubbles meet MPs at grain boundaries; only touching bubbles act	p3: Raman, confocal, size series	Partial: co-location is expected from freeze exclusion	Fig. 2c; cryo-SEM, NanoSIMS ^{18}O
S4	Oxidized MPs refine ice → more bubbles → faster oxidation	p1C, p4	Correlational; kinetics decelerate	Fig. 2d, 2e, 2f
S5	Chain scission to small molecules	p5: GPC, 13C NMR, GC-MS	Partial: PS only; no mass balance	Fig. 2g
S6	Operates in natural ice and reaches water	—	**Missing**	Fig. 2h, 2i, 2j

1.3 How the two planning figures map onto an eventual five-figure manuscript

Main figure	Purpose	Built from
1 Phenomenon and concept	Ice oxidizes C–H polymers, water does not; the ratchet idea	Fig. 1a, 1c (cleaned O/C), 1h
2 The oxidant	What makes the ROS, and is it really a bubble	Fig. 2a–c plus species probes
3 The feedback	Causal test, ratchet signature, bound at realistic loadings	Fig. 1f–g, 2d–f
4 Fate	Chain scission, products, ^{13}C budget, nanoplastic release	Fig. 1i, 2g
5 Natural ice and water	Field dose–response and meltwater first flush	Fig. 2h–j

2 True novelty and physics insight

2.1 Not new (reviewers will know these)

- Ice accelerates oxidation of PS particles at −20 °C, assigned to $^1O_2$ in the liquid layer between ice crystals: Tian et al., *Angew. Chem.* 2022.
- Freezing generates bulk nanobubbles; size and number depend on cooling rate and dissolved gas: Qi et al., *Langmuir* 2023.
- Freeze–thaw cycles oxidize MPs in cold soils via $^1O_2$ and H_2O_2; in the field only aromatic polymers oxidized, with no $^1O_2$ for PE or PA: *ES&T* 2025.
- Freeze-concentration accelerates reactions in ice (e.g., nitrite oxidation, Takenaka et al., *Nature* 1992).

2.2 New

- **Two-way coupling between pollutant chemistry and ice microstructure.** Oxidized MPs act like ice-recrystallization inhibitors. This parallels graphene oxide, whose IRI activity depends on its oxidation degree, and it builds on the group's own IRI expertise (He et al., *Acc. Chem. Res.* 2018). This is the contribution to lead with.
- **A polymer-independent ROS source.** If ROS appear in ice with no plastic present (deck p2D/E suggests so), the Angew mechanism, where the excited polymer is the sensitizer, is falsified. This is the cleanest differentiator. Show it with data; do not assert that the Angew EPR was wrong.
- **Molecular-level degradation evidence** (Mw drop, products). This goes beyond Qian et al. (*Nature* 2026: ROS and biological effects, not polymer breakdown). It does not exceed Gao et al. (*Nature* 2026: quantified diacid yields from molten plastic–water microdroplets).

2.3 Physics the paper should make explicit

- **Gas exsolution and heterogeneous nucleation.** Ice rejects dissolved gas, so veins become supersaturated, and hydrophobic MPs lower the bubble-nucleation barrier. This gives a physical reason for bubble–MP contact beyond co-exclusion. Note that the measured size trend (O/C rising from 100 nm to 10 µm) is the opposite of ice-front engulfment theory and needs explaining.
- **Nanobubbles are reaction sites, not O_2 reservoirs.** Inside a 130 nm bubble the pressure is ≈23 bar, so local O_2 is ≈0.2 M. But 3×10^8 bubbles mL^{-1} hold only $\approx10^{-7}$ M O_2 in total, about 10^{-4} of the dissolved O_2. The paper must say what turns the bubbles over and what supplies the 0.98 eV needed to make $^1O_2$.
- **The light problem.** The schematic says "light" converts $^3O_2$ to $^1O_2$, but a −20 °C freezer is dark and ground-state O_2 barely absorbs (spin-forbidden). Run dark and defined-light controls. If light is required, this becomes ice photochemistry, which has a large prior literature on sensitizers concentrated in liquid-like regions.
- **$^1O_2$ chemistry.** $^1O_2$ reacts by ene, [4+2] and heteroatom oxidation. It does not abstract H from saturated C–H at meaningful rates, so PE/PP oxidation needs •OH or radical chemistry. The Fig. 5A mechanism in the deck has this problem.
- **Diffusion length.** $^1O_2$ lives ≈3.5 µs in water and travels ≈150 nm, a quantitative reason why only touching bubbles matter (Fig. 1a).
- **Hidden negative feedback.** Oxidation hydrophilizes the surface, which should reduce bubble nucleation and plastic–water interfacial ROS (Qian et al. show these scale with hydrophobicity). This can explain the levelling-off kinetics (Fig. 1h) and is one of the three outcomes tested in Fig. 2e.
- **Zener bound.** Particle pinning limits grain size to d = 4r/3f. Holding 20–200 µm grains needs MP volume fractions of 10^{-3}–10^{-2}; natural ice is $\leq10^{-7}$ (Fig. 2f). Any field-relevant feedback must therefore act through something more potent, such as dissolved oxidized oligomers acting like PVA.

- **Where bubbles live.** Boundary area scales as d^{-1} and triple-junction length as d^{-2}; the deck's data scale as $d^{-0.6}$ with a step near 100 µm (Fig. 1g).
- **Liquid fraction at −20 °C is set by solutes.** In near-pure water the veins are nanometre-scale premelted films that cannot host 130 nm bubbles or 10 µm particles. State the composition and compute the liquid fraction from the phase diagram.

3 Where the work stands in the field

The work sits where two literatures meet. One is interfacial ROS chemistry, which is hot and contested: plastic–water (Qian, *Nature* 2026), oil–water (Gao, *Nature* 2026; *JACS* 2026) and air–water. At the air–water interface spontaneous H_2O_2 is disputed, bulk nanobubbles themselves are contested, and whether they make •OH at all is doubtful. The other is freeze-accelerated chemistry, which is mature, alongside a crowded microplastic-fate field. Because the proposed mechanism takes on the controversy from both sides, the bar for proof is high.

3.1 Rival ROS sources the paper must rule out

	Hypothesis	What makes the ROS	Discriminating test (Fig. 2a)
H1	Polymer photosensitizer (Tian, *Angew.* 2022)	Excited polymer + O_2	ROS in ice with no plastic; PE vs PS rates; dark
H2	Freezing (Workman–Reynolds) potential (Song, *ES&T* 2024)	Ice–liquid interface field	Salt identity and concentration series
H3	Plastic–water interface •OH, freeze-concentrated (Qian, *Nature* 2026)	Hydrophobic plastic surface	•OH probes; TBA; no plastic
H4	Freeze-exsolved nanobubbles (this deck)	Gas–water interface	O_2 only, no N_2 (fewer bubbles, same pO_2); pressure; D_2O
H5	Contact electrification (PFA freeze–thaw H_2O_2, 2026)	Vial or particle contact	Glass vs PFA vs PP vials; TBA

3.2 Performance against peers

Work	Conditions	Headline result	Depth of evidence
This deck	−20 °C, ~350 h, 9 polymers	PS O/C 0.015 → ~0.2; PS Mw −52%; PE/PP/PVC Mw −7 to −18%; ~85% of products are 2-carbon	XPS, GPC, 13C NMR, GC-MS; no mass balance
Tian, *Angew.* 2022	−20 °C ice, PS	"Competitive with most artificial technologies"; reviews cite ~12 d (one review 48 d) for 500 nm PS	Abstract only checked
Fe(II)/frozen, *Env. Sci. Nano* 2023	Anoxic Fe(II), frozen	14.6× faster than water; Fe(IV), •OH	Mechanistic
***ES&T* 2025, cold soils**	122-day field + lab	Only aromatic MPs oxidized	Field validation
Qian, *Nature* 2026	Dark, room temperature	•OH ≥0.1–1 nM min^{-1}; methyl orange ~90% in 10 d	ROS + biology; no polymer breakdown
Gao, *Nature* 2026	Molten plastic in water	~69% diacid yield from PE/PP	Quantified yields

Verdict: no rate advantage over Angew 2022 (weeks at −20 °C). The deck wins on breadth and molecular evidence, but its "all C–H polymers" claim conflicts with the field result for PE. As a treatment technology it is far behind. Sell the mechanism and the feedback, not the performance.

4 Figure 1 | The story as it stands

Logic arc. State the mechanism (a). List the five claims it rests on and rate the evidence for each (b). Give one panel of the deck's own data per claim (c–i), so a reader sees exactly where the chain is strong and where it breaks. S-tags and Harvey balls in each panel repeat the spine.

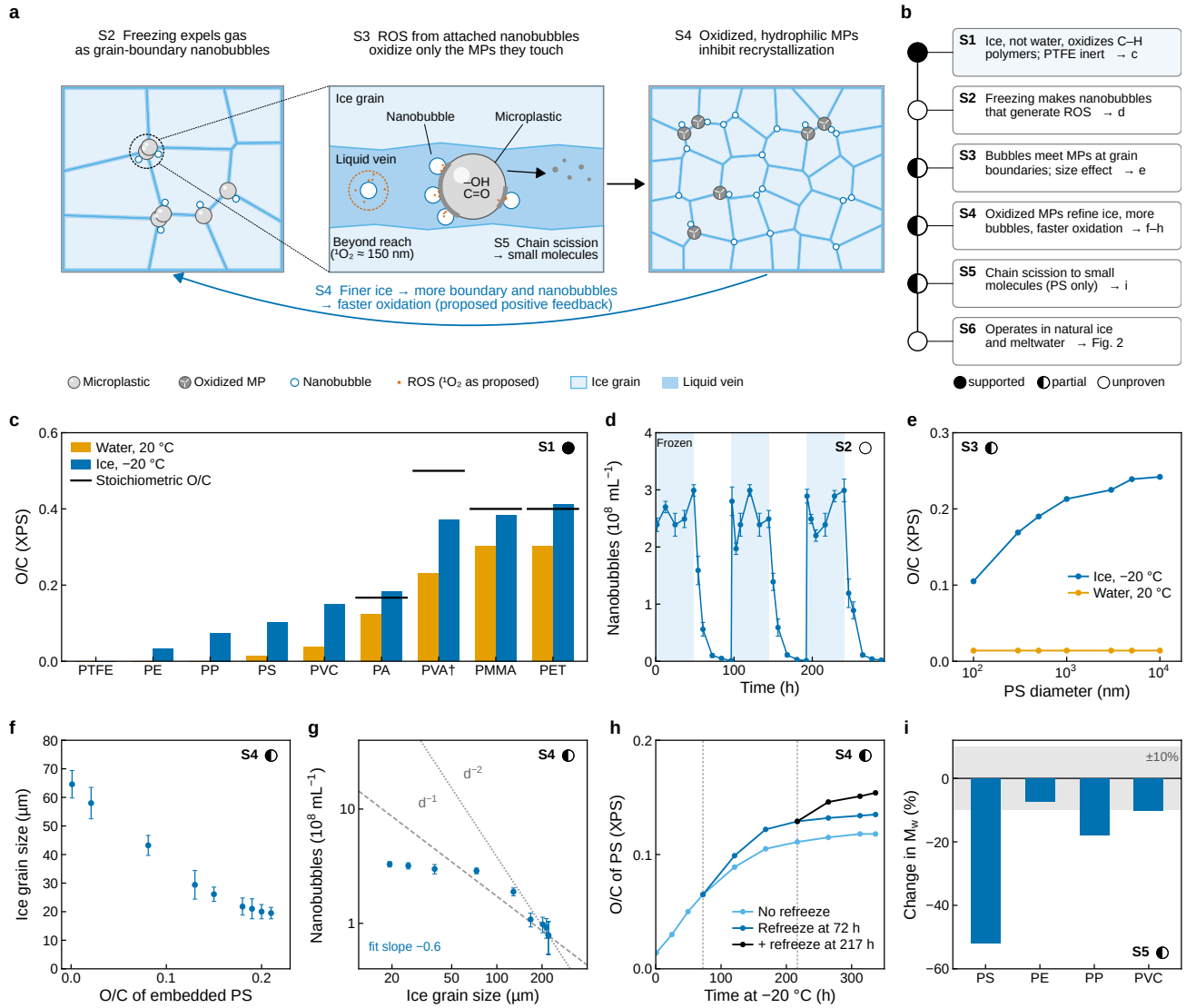


Figure 1 | The story as it stands (deck data, digitized). a, Proposed mechanism. Left: coarse ice with pristine MPs and a few nanobubbles at grain boundaries (S2). Centre: close-up of one liquid vein. Bubbles touching an MP deliver ROS and oxidize its surface (–OH, C=O). A bubble further away than the $^1O_2$ diffusion length (≈150 nm) cannot reach it (S3). Chain scission releases small molecules (S5). Right: oxidized MPs inhibit recrystallization, giving finer ice (S4). Blue arrow: the proposed feedback. **b**, Logic spine. Harvey balls: full = supported, half = partial or correlational, empty = unproven or missing. Arrows point to the panel that carries each step.

c, S1: XPS O/C after incubation in water (20 °C) or ice (−20 °C); black bars, stoichiometric O/C of the repeat unit. †PVA is water-soluble and a known ice-recrystallization inhibitor. **d**, S2: nanobubble count (NTA) through three freeze–melt cycles; shading = frozen. **e**, S3: O/C against PS diameter. **f**, S4, first link: ice grain size against O/C of embedded PS. **g**, S4, second link: bubble density against grain size; grey lines, d^{-1} (boundary area) and d^{-2} (triple junctions); blue, log–log fit. **h**, S4 kinetics: PS O/C at −20 °C with melt–refreeze at 72 h and 217 h (dotted lines). **i**, S5: change in Mw; grey band ±10% (assumed typical GPC run-to-run reproducibility). Values digitized from the deck (±2–5%); error bars as drawn in the deck.

4.1 What each panel means and how it serves the arc

Panel	What it shows	Role in the arc	Editor's reading
a	Mechanism schematic: coarse ice → close-up of one vein → finer ice, with the feedback arrow	The key message in one picture; defines S2–S5	Shows the claim, not the evidence; the light-driven $^1O_2$ step is drawn but undemonstrated
b	Logic spine S1–S6 with Harvey balls	Tells the reader which links hold	Only S1 is solid; S2 and S6 are empty
c	O/C in water vs ice for 9 polymers, with stoichiometric O/C	S1, the phenomenon	PE, PP, PS, PVC show real oxidation. For PA, PMMA and PET the ice values equal the formula O/C, so the "increase" may be loss of adventitious carbon; PVA should be dropped
d	Bubble count reversibly high when frozen, ~0 after melting	S2, that bubbles form	Reproducible, but NTA cannot tell bubbles from nanoplastic fragments; nothing here shows ROS
e	O/C rises with PS size and saturates above 1 µm	S3, that contact matters	Counter to engulfment theory and far shallower than a per-particle nucleation model; needs physics
f	Grain size falls from ~65 to ~20 µm as O/C rises to 0.21	S4, first link (oxidation → ice)	Cleanest panel for the new idea; still needs causality (Fig. 2d)
g	Bubble density vs grain size, log–log, with geometric scalings	S4, second link (ice → bubbles)	Follows neither scaling (slope −0.6, step near 100 µm): location of bubbles unresolved
h	Oxidation kinetics with two melt–refreezes	S4, third link (bubbles → rate)	Every curve slows and levels off; refreeze bumps fit re-oxygenation as well as feedback
i	Mw change for PS, PE, PP, PVC	S5, degradation	Only PS (−52%) exceeds plausible GPC error; surface-limited O/C vs bulk Mw halving must be reconciled

Left out of Fig. 1, on purpose: fluorescence ROS images (DCF and Rh-6G both freeze-concentrate into veins, so co-localization is expected whatever the mechanism); the TEMP–EPR trace (non-specific without D_2O and azide controls); SEM images; the 3-point O/C vs bubble plot (2.9–3.1×10^8 mL^{-1}, within NTA precision). Their replacements are in Fig. 2.

5 Figure 2 | What would make it a *Nature Water* paper

Logic arc. (1) Pin down the oxidant: a–c. (2) Prove the feedback and bound it: d–f. (3) Follow the carbon: g. (4) Show it in natural ice and meltwater: h–j. Every data panel is a **simulated expectation**; each one states the result that supports the story and the result that would falsify it.

a

Manipulation readout: ROS yield and PS oxidation rate in ice	H1 Polymer sensitizer	H2 Freezing potential	H3 Plastic surface	H4 Nano-bubbles	H5 Contact charging
No plastic (ROS probe only)	▼	○	▼	○	○
O_2 only; no N_2 (same pO_2)	○	○	○	▼	○
Dark instead of light	▼	○	○	▼	○
Reversed-potential salt	○	◆	○	○	○
D_2O instead of H_2O	▲	○	○	▲	○
•OH scavenger (TBA)	○	▼	▼	○	▼
Glass instead of PFA vial	○	○	○	○	▼

▼ suppressed ▲ enhanced ◆ altered ○ unchanged H4 = deck mechanism

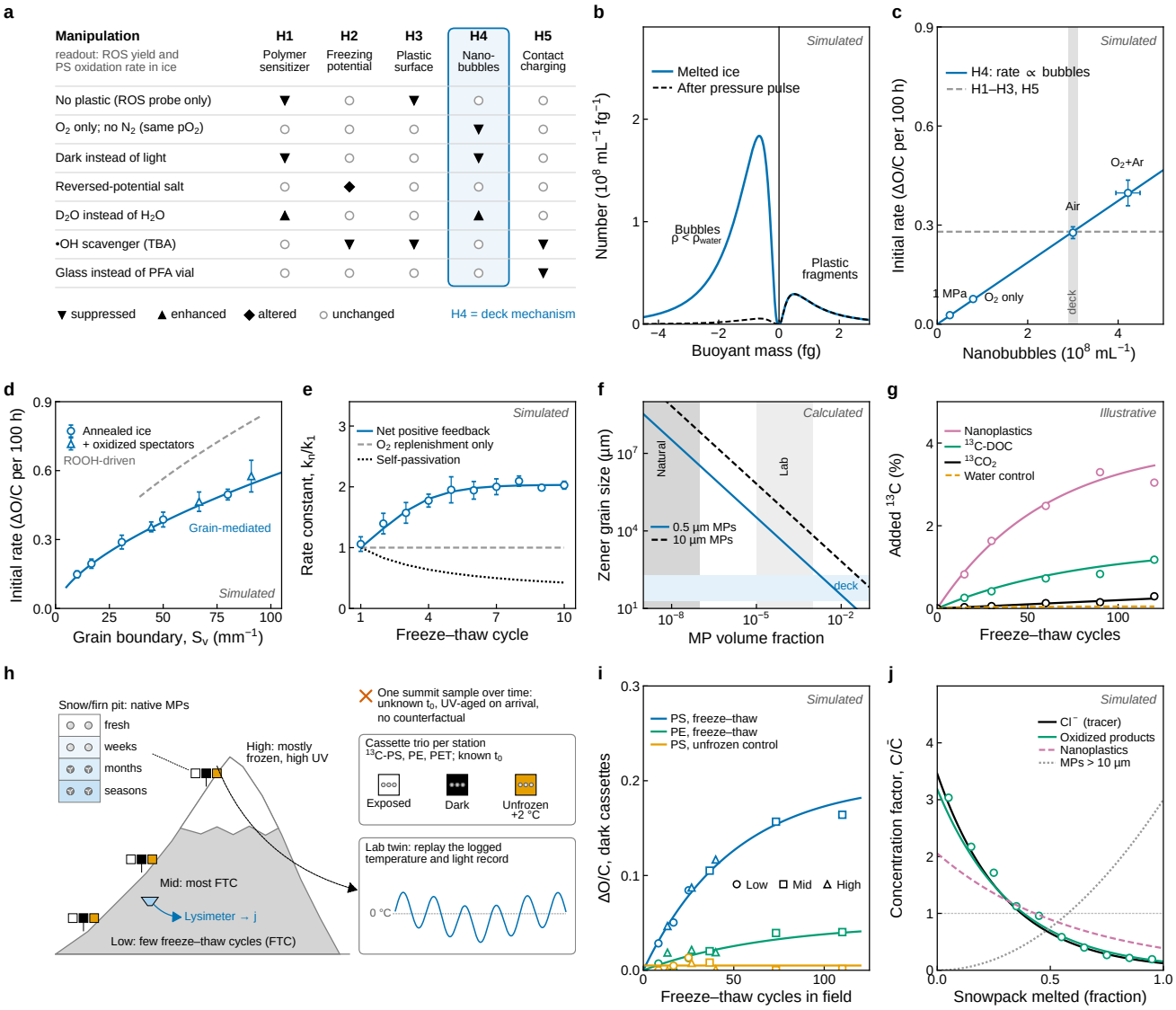


Figure 2 | Decisive experiments and their expected outcomes (simulated). a, Discrimination matrix: five candidate ROS sources against seven manipulations; readout, ROS-probe yield and PS oxidation rate at −20 °C. Each column has a unique signature. H4 is drawn as in the deck (light-driven); if it runs in the dark, the "Dark" cell becomes ○. **b**, Resonant-mass (Archimedes) test: an air bubble has negative buoyant mass (130 nm, −1.15 fg), an oxidized PS fragment positive (300 nm, +1.1 fg); a pressure pulse removes bubbles only. **c**, Oxidation rate against bubble density at fixed pO_2 (1 MPa freezing; O_2 only, no N_2; air; O_2 + Ar); grey band, the deck's 7% range; n = 3, mean ± s.d.

d, Initial oxidation rate of fresh h-PS against grain-boundary density S_v = 2/d, with grain size set by annealing or by pre-oxidized, hydroperoxide-reduced PS-d8 spectators. Collapse on one curve means the feedback is set by the ice; spectators above the curve point to ROOH chemistry. **e**, Ratchet test: per-cycle rate constant normalized to remaining sites, k_n/k_1, over 10 freeze–thaw cycles at fixed O_2, with three outcomes. **f**, Zener-limited grain size d = 4r/3f against MP volume fraction; bands: natural ice/snow (dark), typical lab dosing (light), deck grain sizes (blue). **g**, ^{13}C budget (illustrative magnitudes): nanoplastics, ^{13}C-DOC and $^{13}CO_2$ against freeze–thaw cycles; dashed, water control.

h, Field design: elevation transect bracketing freeze–thaw exposure; a cassette trio of reference MPs at each station (exposed, dark, unfrozen +2 °C); loggers; a snow/firn pit; a lysimeter; a lab twin that replays the logged record. Red ×: why one summit sample measured over time is not enough. **i**, Expected ΔO/C of dark cassettes against cumulative field freeze–thaw cycles; markers = station. **j**, Meltwater first flush: concentration factor against fraction of snowpack melted. Model details in Section 5.3; scripts in *simulate_expectations.py*.

5.1 What each panel means and how it serves the arc

Panel	Experiment	Supports the story if…	Falsifies or reframes if…
a	Seven manipulations × five ROS sources	The response pattern matches column H4	Another column fits; rewrite the mechanism around it (H3 is still a publishable story)
b	Buoyant-mass sign of the frozen-then-melted population	Negative-mass population that a pressure pulse removes	Mostly positive mass: the "bubbles" are particles or nanoplastics
c	Bubble number varied 15× at fixed dissolved O_2	Rate rises in proportion to bubbles	Flat rate: drop nanobubbles as the oxidant
d	Grain size set without changing the chemistry	Both series collapse on one curve, rate ~ $S_v^{0.61}$	Spectators sit above the curve: ROOH autocatalysis, not ice structure
e	10 cycles, fixed O_2, fixed period	k_n/k_1 rises (≈2× by cycle 6)	Flat: re-oxygenation ratchet. Falling: self-passivation. Retitle away from "positive feedback"
f	Calculation, plus refreeze of particle-free filtrate	Refinement persists at ≤10^{-5} via dissolved oligomers	Refinement needs ≥10^{-3}: restrict the feedback claim to high-loading hotspots
g	^{13}C-PS mass balance	Fragments and DOC dominate, mineralization <0.5%: "transformation" framing	Negligible mobile carbon: the degradation claim weakens to surface aging
h	Controlled field deployment	—	—
i	Dark-cassette ΔO/C vs field freeze–thaw count	All stations on one curve; unfrozen controls flat; PE above zero	Station-specific curves: elevation or UV, not freeze–thaw, controls it
j	Lysimeter or snow-column melt fractions	Products track the chloride ionic pulse	No early enrichment: products are not grain-boundary-held

5.2 Why these panels and not others

- Each Fig. 2 panel answers one of the referee's foreseeable objections to Fig. 1 (Section 9), in the order a referee would raise them: is it a bubble, is it the oxidant, is the loop real, does it survive real loadings, where does the carbon go, does it happen outdoors.
- Species probes (FFA, terephthalate, coumarin, DMPO, $^{18}O_2$) are folded into the matrix rather than given their own panels. ^{18}O labelling alone does not separate the routes, because both deliver most carbonyl oxygen from O_2 via autoxidation.
- Temperature (−2 to −20 °C) and matrix (brine, snowmelt, DOM) series belong in the SI. They are essential, but their outcome cannot be predicted well enough to plot honestly.
- Ice-binding proteins and PVA are excluded as grain-size controls because they quench ROS.

5.3 Models and parameters behind the simulated panels

Panel	Model	Parameters (anchors)
b	Lognormal size distributions → buoyant mass $m = (\rho - \rho_w)\pi D^3/6$	Bubbles 130 nm ($\sigma_{\ln d}$ 0.25), 3.0×10^8 mL^{-1} (deck p2B); fragments 300 nm, ρ 1080 kg m^{-3}, 0.5×10^8 mL^{-1}; pulse removes 97% of bubbles
c	H4: rate = $R_{air}\cdot N/N_{air}$; rivals: rate = R_{air}	R_{air} = 0.28 ΔO/C per 100 h (deck p3D initial slope); N = 0.3, 0.8, 3.0, 4.5; noise 10% (x), 8% (y); n = 3
d	rate = $R_{air}(S_v/S_{ref})^{0.61}$	S_{ref} = 2/65 µm (deck p4D); exponent from deck p4E fit; ROOH alternative 1.5×
e	$d\theta/dt = k\cdot N(\theta)(1 - \theta/\theta_{max})e^{-t/\tau}$, pool reset at refreeze	k = 0.0016 h^{-1}, τ = 100 h, θ_{max} = 0.25, 72 h frozen per cycle (reproduce deck p4G: 0.065 at 72 h, plateau 0.118); $N = (d_0/d(\theta))^{0.61}$, $d(\theta) = 5.0 + 61.1e^{-\theta/0.145}$ µm; passivation $N = 1 - 0.75\theta/\theta_{max}$
f	Zener: d = 4r/3f	r = 0.25 and 5 µm; natural ≤10^{-7} (Arctic sea ice ≈1.2×10^4 particles L^{-1} of ~20 µm ≈ 5×10^{-8}); lab 10–1000 mg L^{-1} ≈ 10^{-5}–10^{-3}
g	Saturating release pools (illustrative)	NP $4.0(1 - e^{-n/60})$%, DOC $1.5(1 - e^{-n/80})$%, CO_2 0.002n%, water 0.05%
i	$\Delta O/C = a(1 - e^{-n/n0})$	PS a 0.20, n0 50; PE a 0.05, n0 70; unfrozen 0.005; season totals 25/110/40 cycles; noise σ 0.006
j	Preferential elution $C/C_{mean} = e^{-F/\tau}/[\tau(1 - e^{-1/\tau})]$	τ: chloride 0.30, products 0.33, nanoplastics 0.60; intact MPs 3F^2 (snowpack filtration, late release)

6 Field study design (and why a single summit sample is not enough)

Should we just sample mountain-top ice and track microplastic degradation over time? Not on its own. Summit MPs arrive after atmospheric transport under strong UV, so their starting state is unknown and any aging may predate deposition. Summit ice sees few freeze–thaw cycles, which is the variable the mechanism depends on. There is no counterfactual, counts are tens to hundreds per litre, and contamination dominates. A referee can attribute any trend to UV, deposition history or handling.

Element	Specification	Why
Sites	Three stations on one elevation transect (few / most / few freeze–thaw cycles), optionally a seasonally frozen soil or lake-ice site	Freeze–thaw count becomes the dose variable; elevation and UV are decoupled from it
Reference MPs	^{13}C-PS (500 nm, 10 µm), PE, PET; fully characterized at t_0	Known starting state; ^{13}C traces fate against background plastic
Containment	Sealed borosilicate cells filled with site snow or meltwater, vented through stainless frits; no fluoropolymer or PP parts	Nanoparticles must not escape; avoids H5 artefacts and contamination
Treatments	Exposed / dark / unfrozen (+2 °C, thermostatted); n ≥ 3 per treatment per retrieval	Separates photo-oxidation and freezing; the unfrozen cell is the counterfactual
Retrievals	1/3, 2/3 and end of season	Gives a dose–response, not a single endpoint
Loggers	Temperature in each cell (±0.1 °C, 10 min); UV-A/B and PAR	Count cycles with a −1/+1 °C hysteresis band, so sensor noise is not counted
Native archive	Snow/firn pit at the high station; fresh layer = atmospheric baseline	Secondary evidence on real MPs; never the primary test
Meltwater	Lysimeter on unseeded snow for native material; lab snow columns with natural snow for ^{13}C-PS	Do not seed open snowpack with plastics; permits and full recovery otherwise
Lab twin	Climate chamber replays each station's logged T(t) and light	Separates climate from site factors (deposition, biology)
QA	Field, travel and procedural blanks; glass and metal tools; cotton clothing	MP studies fail on contamination
Readouts	ΔO/C, C1s fitting, FTIR carbonyl index, GPC, nanoplastic counts, ^{13}C-DOC, $^{13}CO_2$	Same metrics as the lab, so field and lab plot on one axis

One winter season (November–May) with the lab twin running in parallel is enough to produce Fig. 2i and 2j.

7 Prioritized plan with decision gates

#	Experiment	Time	Decision gate
P1	Bubble identity (2b) and dose–response at fixed O_2 (2c)	2–3 wk	If oxidation does not track bubble number, drop H4 and reframe as ice-confined interfacial ROS (H2/H3)
P2	Discrimination matrix (2a) with species probes, D_2O, azide, TBA, dark/light, vial material	3–4 wk	Species identity sets the mechanism language; if •OH, remove $^1O_2$ from the schematic
P3	Feedback causality (2d) and ratchet (2e)	4–6 wk	If flat or falling, retitle: "re-oxygenation ratchet" or "self-limiting" instead of "positive feedback"
P4	Realistic loadings and matrices (2f; brine, snowmelt, DOM; −2 to −20 °C)	3 wk	If refinement vanishes at ≤10^{-5}, confine the feedback claim to hotspots
P5	^{13}C budget (2g); quantified products with standards; nanoplastic release	6–8 wk	Decides between "degradation" and "surface aging" language
P6	Field season, lab twin, meltwater (2h–j)	One winter	Required for *Nature Water*

P1–P3 (plus partial P5) are enough for *Nat. Commun.*, *JACS* or *Angew. Nature Water* needs P1–P6.

8 Journal fit and likelihood (subjective editorial estimates)

Journal	As the deck stands	With the decisive package	Comment
Nature Water	~5% (likely desk reject)	15–25%	Needs the water-systems "so what": field plus meltwater
Nature Sustainability	<3%	~5%	Poor fit either way: weeks at −20 °C is not a sustainable treatment
Nature Communications	15–20%	35–45%	Most realistic Nature-family home
JACS	10–15%	30–40%	Mechanism must be nailed: isotopes, kinetics, species
Angew. Chem.	25–30%	45–55%	Published Tian 2022, so a mechanistic correction fits
PNAS* / *Sci. Adv.	10–20%	25–35%	
ES&T	40–50%	65–75%	Safe home for the environmental-fate version

Strategy: send *Nature Water* a presubmission enquiry built around the feedback concept and the Zener sanity check (Fig. 2f). If interest is lukewarm, target *Nat. Commun.* after P1–P3.

9 Referee checklist (what a picky reviewer will ask)

- **Nanobubble identity:** NTA counts nanoplastics too; the Rh-6G "nanobubble" channel is a continuous vein network (freeze-concentrated dye). Needed: resonant mass, degassed/Ar controls, pressure collapse, cryo-EM.
- **ROS identity:** TEMP → TEMPO and DCFH are non-specific; DCFH photo-oxidizes under the imaging laser. Needed: D_2O enhancement, azide/FFA quenching, 1270 nm emission, •OH probes and TBA.
- **Light:** state the light conditions of every experiment; freezers are dark.
- **O/C metric:** report ΔO/C against pristine, fitted C1s components and FTIR carbonyl index; PS O/C under nominally the same condition reads 0.10 (p1B), 0.12 (p4G) and 0.19–0.21 (p3D/E). Specify size and time.
- **Surface vs bulk:** XPS probes ~5–10 nm, but PS Mw halves across 0.5–1 µm particles. Show depth profiles and GPC mass recovery.
- **Products:** m/z 105/77/51 is a generic benzoyl pattern. Identify with authentic standards, quantify yields, close the mass balance.
- **Feedback:** correlation is not causation; the plateau contradicts autocatalysis; melt–refreeze confounds re-oxygenation.
- **Relevance:** loading, solution composition and liquid fraction, temperature range, container material (PFA), natural matrices, field data.
- **Tone:** demonstrate rather than assert that prior work (Angew 2022 EPR) was wrong; its authors may review.

10 Provenance and caveats

- Fig. 1 values are digitized from the deck's figures (±2–5%). PE, PP and PVC Mw are read off a log axis.
- Fig. 2 data panels are simulated from the models in Section 5.3 and are labelled "Simulated", "Calculated" or "Illustrative". Fig. 2g magnitudes are guesses; only the shape carries meaning.
- Competitor details come from abstracts, indexes and press coverage. Publisher full texts (nature.com, Wiley) could not be opened from the analysis environment, so verify numbers before citing them.
- Journal likelihoods are subjective editorial estimates, not statistics.
- Figures were rendered in Liberation Sans (metric-compatible with Arial); the scripts use Arial automatically where it is installed.

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Appendix Files and reproduction

File	Content
fig1_current_spine.py / fig2_roadmap.py	One script per figure; vector PDF + SVG + PNG to out/
simulate_expectations.py	All Fig. 2 models; writes data/simulated/*.csv
data/deck/*.csv	Digitized deck values with source page and panel in each header
common.py	Nature-style preset, mm-grid helpers, schematic primitives
build_report.py	This report